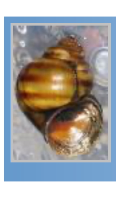


Callinina georgiana (banded mysterysnail)
Mollusks-Gastropods
Native Transplant

Collection Info
Point Map
Species Profile
Animated Map
Impacts

Jason M. Crockwell

Callinina georgiana (I. Lea, 1834)

Common name: banded mysterysnail

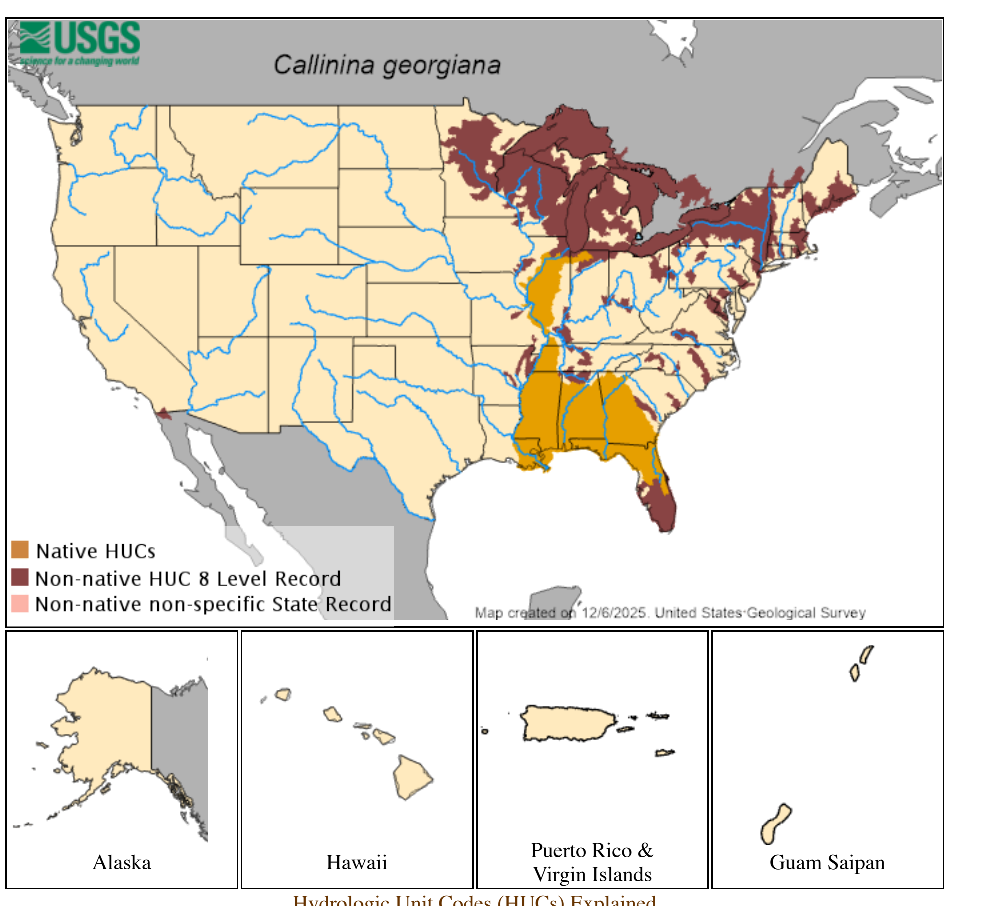
Synonyms and Other Names: Viviparus georgianus, banded applesnail, pondsnail, *Vivipara contectoides* Binney 1865, *Paludina georgiana* I. Lea, 1834

Taxonomy: available through ITIS

Identification: *Callinina georgiana* is a freshwater prosobranch (gills in front of heart) snail species complex with a thin and smooth shell, yellow-green in color with a straight outer lip, often with four distinctive brown bands present on the body whorl (Clench 1962; Mackie et al. 1980). The species complex has a very variable shell morphology, and the shell bands are sometimes absent (Clench and Fuller 1965), but it always has an adextral (right-handed) shell with 3–5 inflated whorls separated by deeply indented incisions. *Callinina* can be distinguished from the genus *Vivaparus* and other genera within the family Viviparidae by there concentric operculum which is characterized by a difference in color between the marginal and enamel regions, a small and smooth nuclear region, and a more narrow inner enamel region (Zhang and von Rintelen 2021).

Size: One-year old snails are 12–17 mm; at 2 years, 17–21 mm; and at 3 years, 21–30 mm (Lee et al. 2002). The maximum height is 45 mm (Jokinen 1992).

Native Range: The banded mysterysnail is native to North America, and is generally found in waterbodies of the southeastern and midwestern United States, from Central Florida up to northern Illinois, and throughout the eastern part of the Mississippi Drainage (Clench 1962). It is unclear whether the native range of this species includes the Tennessee River Drainage, but it is likely introduced to the drainage given the absence of the species from very extensive surveys from shell collectors in the area during mid-late 1800s (Clench 1962).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences: It was first introduced into the Hudson River basin of New York in 1854 (Strayer, 1987), but the population failed. More individuals were released in 1867, resulting in an established population in the Hudson Drainage (Clench,1962; Strayer, 1987). The species was historically absent from most of the Atlantic coast drainages, and is known to have become established in the northeastern and midwestern United States as far back as the early 1900's due to intentional releases, many from the aquarium trade (Clench, 1962; Mills et al., 1993; Dillon et al., 2006).

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Callinina georgiana* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AL	1903	2005	2	Pickwick Lake; Wheeler Lake
AR	1914	2004	3	Bayou Meto; Cache; Lower St. Francis
CA	1995	1995	1	Cottonwood-Tijuana
CT	1962	1983	1	Housatonic
DC	1962	1962	1	Middle Potomac-Anacostia-Occoquan
FL	1905	2019	15	Big Cypress Swamp; Caloosahatchee; Cape Canaveral; Everglades; Florida Southeast Coast; Hillsborough; Kissimmee; Lake Okeechobee; Myakka; Northern Okeechobee Inflow; Peace; Sarasota Bay; Tampa Bay; Vero Beach; Western Okeechobee Inflow
GA	1969	1970	2	Lower Ogeechee; Upper Ogeechee
IL	1906	2021	6	Chicago; Des Plaines; Little Calumet-Galien; Lower Fox; Lower Ohio-Bay; Pike-Root
IN	1927	2014	5	Lower Wabash; St. Joseph; St. Joseph; Tippecanoe; Wabash
IA	1967	2006	2	Coon-Yellow; Flint-Henderson
KY	1927	1999	3	Lower Cumberland; Middle Ohio-Laughery; Tradewater
ME	1907	2022	5	Lower Androscoggin River; Maine Coastal; Penobscot River; Presumpscot; St. George-Sheepscot
MD	1962	2013	3	Gunpowder-Patapsco; Middle Potomac-Anacostia-Occoquan; Middle Potomac-Catoctin
MA	1916	2019	7	Charles; Concord River; Housatonic; Narragansett; Nashua River; Outlet Connecticut River; Westfield River
MI	1980	2025	32	Au Gres-Rifle; Au Sable; Boardman-Charlevoix; Brule; Carp-Pine; Cheboygan; Clinton; Detroit; Fishdam-Sturgeon; Flint; Keweenaw Peninsula; Lake Huron; Lake Michigan; Lake Superior; Lower Grand; Manistee; Manistique River; Maple; Menominee; Michigamme; Muskegon; Ontonagon; Ottawa-Stony; Pere Marquette-White; Raisin; Saginaw; Shiawassee; St. Joseph; St. Marys; Thornapple; Tiffin; Tittabawassee
MN	1930	2018	25	Beartrap-Nemadji; Big Fork; Clearwater; Clearwater-Elk; Cloquet; Coon-Yellow; Crow; Crow Wing; Eastern Wild Rice; Elk-Nokasippi; Kettle; Leech Lake; Little Fork; Long Prairie; Lower Minnesota; Mississippi Headwaters; Otter Tail; Pine; Platte-Spunk; Prairie-Willow; Red Lakes; Redeye; Rum; Sauk; Twin Cities
MO	1936	2013	1	Bourbeuse
NH	2017	2022	1	Merrimack River
NJ	1912	2018	3	Hackensack-Passaic; Lower Delaware; Rondout
NY	1854	2024	39	Ausable River; Black; Buffalo-Eighteenmile; Cattaraugus; Chenango; Conewango; French; Housatonic; Hudson-Hoosic; Hudson-Wappinger; Indian; Irondequoit-Ninemile; Lake Champlain; Lake Erie; Lake Ontario; Lower Genesee; Lower Hudson; Mettawee River; Middle Hudson; Mohawk; Niagara River; Northeastern Lake Ontario; Oak Orchard-Twelvemile; Oneida; Oswegatchie; Oswego; Raquette; Sacandaga; Salmon-Sandy; Sandy Hook-Staten Island; Saranac River; Schoharie; Seneca; Southwestern Lake Ontario; St. Lawrence; St. Regis; Upper Genesee; Upper Hudson; Upper Susquehanna
NC	2009	2022	5	Middle Roanoke; Northeast Cape Fear; Upper Catawba; Upper Neuse; Upper Pee Dee
OH	1962	2014	4	Cuyahoga; Middle Ohio-Laughery; Ohio Brush-Whiteoak; Tuscarawas
PA	1962	2024	5	Conemaugh; French; Schuylkill; Sinnemahoning; Upper Ohio
SC	1995	2024	2	Cooper; Lake Marion
TN	1962	2015	3	Lower Cumberland-Sycamore; Stones; Upper Duck
VT	1962	2025	5	Lake Champlain; Lamoille River; Mettawee River; Otter Creek; Richelieu
VA	1949	2017	4	Lower Potomac; Middle Potomac-Anacostia-Occoquan; Middle Roanoke; Upper Roanoke
WI	1906	2019	34	Bad-Montreal; Baraboo; Beartrap-Nemadji; Black; Black-Presque Isle; Brule; Buffalo-Whitewater; Castle Rock; Door-Kewaunee; Duck-Pensaukee; Flambeau; Jump; Lake Dubay; Lake Michigan; Lake Winnebago; Lower Chippewa; Lower St. Croix; Manitowoc-Sheboygan; Menominee; Middle Rock; Milwaukee; Namekagon; Oconto; Ontonagon; Peshtigo; Red Cedar; South Fork Flambeau; Upper Chippewa; Upper Fox; Upper Fox; Upper Rock; Upper St. Croix; Upper Wisconsin; Wolf

Table last updated 12/8/2025

† Populations may not be currently present.

Ecology: This species is found in freshwater low-flow lentic streams, lakes, and ponds. It is often present with soft, silty and/or rocky substrates, but is present in a variety of habitats, including sand and detritus bottoms (Duch 1976; Browne 1978). It is usually absent from larger, faster-flowing rivers (Katoh and Foltz 1994); however, it can survive conditions of high water velocity in the St. Lawrence River, and may even be better adapted than the introduced *Bithynia tentaculata* (mud bithynia) to such habitat (Vincent 1979). Individuals are generally found in waters with pH between 6.3 and 8.5. (Duch 1976; Wade and Vasey 1976; Vincent 1979; Jokinen and Pondick 1981; Pace and Szuch 1985; Jokinen 1992; Lee et al. 2002). However, shell repair is reduced at lower pHs (David et al. 2020). Banded mysterysnail is also sensitive to high salt concentrations. As a proxy for road salt pollution, experimental tanks spiked with chloride (15, 100, 250, 500, and 1000 Cl–/L) led to significant mortality of banded mysterysnails at concentrations >500 mg Cl–/L (Hintz et al. 2017).

Banded mysterysnail often lives at high densities, sometimes up to 864/m2 (Pace and Szuch 1985; Lee et al. 2002). It inhabits shallow waters, often amongst macrophytes, in spring to fall, before moving out to deeper areas to overwinter away from shore (Jokinen et al. 1982; Wade 1985; Lee et al. 2002), where it will burrow under the substrate for a period of inactivity (Pace and Szuch, 1985). In more open waters, fall migration begins earlier than in smaller lakes and ponds (Lee et al. 2002). Most growth generally occurs when waters become warmer in spring and summer, although reduced growth continues in winter (Browne 1978; Jokinen et al. 1982).

Breeding takes place in the spring (Pace and Szuch 1985). It is dioecious (distinctly male or female) and ovoviviparous, with females laying eggs singly in albumen-filled capsules and brooding them for 9–10 months; this species is one of very few gastropods to give birth to live young (Browne 1978; Jokinen et al. 1982; Rivest and Vanderpool 1986; Lee et al. 2002). In its non-native range, fecundity ranges from 4–81 young per female per year, but on average, is closer to 11 young per female (Vail 1978; Jokinen 1992; Keller et al. 2007). Females can brood more than one clutch of young at a time and the number of young in one brood is positively related to the size of the female (Vail 1977). Some populations are known to reach sexual maturity within one or two years and reproduce iteroparously (more than once in life), while other populations have been known to breed semelparously (breed only once in life), not reaching sexual maturity until year three of life before dying (Dillon et al. 2006). The lifespan of the female banded mysterysnail is typically between 28–48 months, while males live between 18–36 months (Jokinen et al. 1982; Lee et al. 2002). The group is sexually dimorphic with females growing larger and faster than males, and reproductive females usually larger than 16 mm (Browne 1978; Buckley 1986).

Banded mysterysnail is known to be a facultative, or even obligate, filter-feeding detritivore (Browne 1978; Lee et al., 2002). This species grazes on diatom clusters found on silt and mud substrates, but may require the ingestion of some grit to break down algae (Duch, 1976). With a variable diet, it will readily consume a herbivorous diet of algae and diatoms, but will also consume fish eggs (Duch 1976; Eckblad and Shealy 1972; Mackie et al. 1980; Jokinen et al. 1982; Lee et al. 2002).

Means of Introduction: The earliest introduction of this species to the Hudson River drainage was made by an amateur conchologist who purposefully released around 200 of these snails simultaneously into the river in the 1850s (Jokinen 1992; Mills et al. 1993). Later introductions were likely made via release from aquaria (Mills et al. 1993), but a one study found that this species is very resistant to desiccation, making it very capable of being dispersed over land via boat or other means (Havel et al. 2014).

Great Lakes: The first record of this species in the Great Lakes basin most likely came from the Hudson River drainage, via the Erie Canal and Mohawk River, in 1867 (Mills et al. 1993). It was later reported from the Lake Michigan watershed by 1906 and Lake Erie by 1914.

Status: This species is considered established in the waterbodies in which it is introduced.

Great Lakes: Widespread, with populations reproducing and overwintering at self-sustaining levels in all five Great Lakes.

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological | **Economic** | **Other**

Callinina georgiana has been shown to significantly reduce survival of largemouth bass eggs in guarded nests both in the laboratory and in ponds, and may contribute to high incubation mortality seen in natural field settings (Eckblad and Shealy, 1972).

Various fish and bird species are known to feed on the snail (Eckblad and Shealy, 1972; Smith, 2007).

This species is known to be the intermediate host for trematodes and has, as a result, been involved in spreading parasites to aquatic birds, resulting in large avian die-offs. In 2007, over 3,000 scaup and coots died in a Northern Wisconsin lake as a result of ingesting the infected, non-native snails, with many more birds unable to fly because of the infection. (Smith, 2007).

This species has been documented in high densities where present, and to be more successful in the north, further from its known native range (Dillon et al., 2006). It is often the dominant member of the macrofauna in its trophic level, both in number and function (Browne, 1978).

Because some populations of the banded mysterysnail are semelparous (dying off after one breeding event), this can create a large concentration of dead snails in habitats and leave behind significant shell debris (Dillon et al., 2006).

Remarks: In 2021, the Freshwater Mollusk Conservation Society completed a petitoned name change to *Cassinina georginia* from *Viviparus georgianus* based on the work of Stelbrink et al. 2020 (Freshwater Mollusk Conservation Society 2021; Stelbrink et al. 2020).

This species is very similar to the European *Viviparus viviparus*. It is possible that some introduced populations could actually be *V. viviparus* (Mills et al. 1993). This species is also similar in shell shape and distribution with *Viviparus intertextus* and *Viviparus subpureus* (K. Cummings, Illinois Natural History Survey, pers. comm., July 24, 2018).

Using allozyme data, Katoh and Foltz (1994) found that *Callinina georgiana* is actually a species complex; speciation has occurred within the group in the southeastern United States due to the separation of populations by large rivers that act as barriers for dispersal. Three distinct species were found to be in the Georgia-Florida drainages, each grouping by drainage: *C. georgiana* formed a western group in the Choctawhatchee and Apalachicola River Drainages, Callinina limi formed a central group in the Ochlockonee River Drainage and southwestern Georgia, while *Callinina goodrichi* was found to be present in the most eastern rivers extending into the Florida Peninsula. The genetic identities of some populations remain undetermined, such as those of the Altamaha, Mississippi and St. Lawrence River drainages, and are therefore named as part of the *C. georgiana* species complex (Katoh and Foltz 1994). A later genetic study found populations introduced in New York to group with the western complex, *Callinina georgiana* (David et al. 2017).

This species' migration, which typically results in individuals burrowing under mud during the fall and winter months, has led to an underrepresentation of the species during sampling (Pace and Szuch 1985).

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Other Resources:
GLIFWC-Maps

Great Lakes Waterlife

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